

Name: Weiwei Xu (徐伟伟)**ORCID:** 0000-0002-9587-6683**Email:** wwxu@pku.edu.cn**Homepage:** wwxu.github.com**Institution:** National Astronomical Observatories, Chinese Academy of Sciences (NAOC)**Address:** 20A Datun Road, Chaoyang District, 100101, Beijing, China**EDUCATION AND JOB:**

- May 2022 - Now: Assistant researcher, NAOC, Beijing, China.
Collaborator: Prof. Ran Li
- May 2019 - Apr 2022: Post-doc, Kavli Institute for Astronomy and Astrophysics (KIAA), Peking University, Beijing, China.
Collaborated advisor: Prof. Linhua Jiang
- Jul. 2016 - Nov. 2018: visiting PhD student, the Argelander-Institut für Astronomie (AIfA), University of Bonn, Bonn, Germany.
Supervisor: Prof. Thomas Reiprich
- Sep. 2014 - Jan. 2019: PhD in astrophysics, NAOC, Beijing, China.
Supervisor: Prof. Lijun Gou
- Sep. 2011 - Jun. 2014: Master in Astrophysics, NAOC, Beijing, China.
Supervisor: Prof. Jinlin Han
- Sep. 2007 - Jun. 2011: Bachelor in applied physics, Shandong University, Weihai, China

RESEARCH INTEREST:

- galaxy clusters and cosmology, weak lensing, data analysis, optical & X-ray astronomy

SKILL:

- AI coding, Python, MPI, spectroscopic and photometric data, heasoft and Xspec software

ACADEMIC SERVICE:

- Reviewed for Journal: Astronomy & Astrophysics (A&A)
- Reviewed for Chandra observation proposal

RESEARCH FUNDING:

- NSFC young scientist program, 300k RMB, PI, 2023-2025, “the identification of very extended galaxy clusters with XMM-COSMOS and CDF-S data”
- National Key R&D Program, 4800k RMB, co-I, 2022-2027, “Conducting precision measurements of the dark matter universe using CSST data”

ACADEMIC PUBLICATION:

- **As of Jun. 2026, 18 published papers in total, 208 citations. H-index: 9**
7 first-author and 1 corresponding-author papers published;
10 co-authored published papers published.
- **Published papers as first/corresponding author:**
 8. Chen, X., **Xu, W.***, Li, R., et al. 2025, apj, 990, 171, *A Comparative Study of Halo Mass Estimates from Group Catalogs and Lensing Signals*. *: 通讯作者,
 7. **Xu, W.**, Jiang L., et al. , 2024, a&a, 691, A300, *XMM-SERVS X-ray eXtended Galaxy Cluster (XVXGC) catalog*
 6. **Xu, W.**, Shan, H., et al. 2024, apj, 971, 157, *The Measurement of the Splash-back Radius of Dark Matter Halos*
 5. **Xu, W.**, et al., 2022, a&a, 658, A59, *Catalog of X-ray-selected extended galaxy clusters from the ROSAT All-Sky Survey (RXGCC)*

4. **Xu, W.**, et al. 2021, *apj*, 922, 162, *Halo Mass-Concentration Relation at High-Mass End*
3. **Xu, W.**, Ramos-Ceja, M. E., et al. 2018, *a&a*, a162, 17, 第一作者, *A New X-ray Selected Sample of Very Extended Galaxy Groups from the ROSAT All-Sky Survey*
2. **Xu, W.**, Liu, Z., et al. 2016, *mnras*, 455, L26, 第一作者, *X-ray fluorescent lines from the Compton-thick AGN in M51*
1. **Xu, W.**, Wen, Z., et al. 2014, *Science China Physics, Mechanics, and Astronomy*, 57, 2168, 第一作者, *An efficient method to identify galaxy clusters by using SuperCOSMOS, 2MASS and WISE data*

• **Published papers as co-author:**

11. **【accepted】** Cheng, C., Zhang, C., Ibar, E., Smith, R., **Xu, W.**, et al., accepted by *ApJ*, *Beyond the Virial Radius: FAST view of HI in Galaxy Clusters to 5xR200*
10. Qi, L., Zhang, J., **Xu, W.**, et al., 2026, *apj*, 1002, 145, *Multi-wavelength analysis of PSR J0437–4715 with pulse profile modeling*
9. Xie, Y., Chen, X., [...], **Xu, W.**, et al., 2026, *RAA*, 26, 024007, *Mock Observations for the CSST Mission: Multi-Channel Imager–The Cluster Field*
8. Yuan, Z., Wen, Z., **Xu, W.** & Han, J., 2025, *mnras*, 543, 3698, *A catalogue of 3403 galaxy clusters derived from XMM-Newton X-ray images*
7. Spinelli, C., Veronica, A., [...], **Xu, W.**, et al. 2025, *a&a*, 700, A220, *XMM-Newton follow-up of a sample of apparent low surface brightness galaxy groups detected in the ROSAT All-Sky Survey*
6. Wang, C., Li, R., Shan, H., **Xu, W.**, Yao, J., Jing, Y., Gao, L., et al., 2024, *mnras*, 528, 2728, *Assessing mass-loss and stellar-to-halo mass ratio of satellite galaxies: a galaxy-galaxy lensing approach utilizing DECaLS DR8 data*
5. Wang, C., Li, R., Zhu, K., Shan, H., **Xu, W.**, Cappellari, M., Gao, L., et al., 2024, *mnras*, 527, 1580, *MaNGA DynPop - IV. Stacked total density profile of galaxy groups and clusters from combining dynamical models of integral-field stellar kinematics and galaxy-galaxy lensing*
4. Chen, K., **Xu, W.**, & Jiang, L., 2022, *apj*, 940, 79, *Spectroscopic Confirmation of Two X-Ray Diffuse and Massive Galaxy Clusters at Low Redshift*
3. Wang, J., **Xu, W.**, et al. 2020, *apj*, 903(2):103, *Ram Pressure Stripping of HI-rich Galaxies Infalling into Massive Clusters*
2. Liu, J., Liu, Y., [...], **Xu, W.**, et al. 2016, *mnras*, 459, L100, *The clumpy torus around type II AGN as revealed by X-ray fluorescent lines*
1. Chen, Z., Gou, L., [...], **Xu, W.**, et al. 2016, *apj*, 825, 45, *The Spin of The Black Hole in the X-ray Binary Nova Muscae 1991*

OBSERVATION PROPOSALS:

- EP - FXT, 30ksec, ID: Cycle3-0015, approved,, 2026, PI
- EP - FXT, 50ksec, ID: Cycle3-0101, approved,, 2026, PI
- EP - FXT, 30ksec, ID: Cycle2-0045, approved, 2025, PI
- EP - FXT, 120ksec, ID: Cycle2-0079, approved, 2025, Co-I
- AstroSat AO-12, 119ksec, ID: A10-065, observed, 2021, PI
- XMM-Newton AO-19, 60ksec, ID:86388, observed, 2019, Co-I
- **【submitted】** Chandra, 50ksec, submitted, 2026, PI

SELECTED TALKS:

- **【Invited talk】** Academic Report of School of Physical Science and Technology, Ningbo University, July 2025, 45min
- **【Invited talk】** Series Academic Reports, State Key Laboratory of Particle Astrophysics, Institute of High Energy Physics, CAS, Apr. 2025, 45min
- **【Invited talk】** NAOC workshop of physics of galaxy clusters, July 2024, 45min
- **【Invited talk】** Tsinghua DOA lunch talk, Nov. 2024, 40min
- **【Invited talk】** Xtalk online seminars, Dec. 2024, 40min
- **【Invited talk】** KIAA lunch talk, Dec. 2018, 45min
- **【Invited talk】** NAOC Seminar, Dec. 2017, 45 min
- **【Invited talk】** NJU Seminar, Dec. 2017, 45 min
- [Conference talk] Annual Meeting of CSST, July 2026, 15min
- [Conference talk] eROSITA & HUBS Scientific Exploitation Workshop 2024, Mar. 2024, 20min
- [Conference talk] The KooGiG-Junior forum (KIAA Forum on Gas in Galaxies for Early Career Scientists), online, Nov. 2021, 15min
- [Conference talk] 2020 Jing-Guang-Xia Astrophysics Meeting, Xiamen, Dec. 2020, 10min
- [Conference talk] Annual Meeting of Chinese Astronomical Society, Delingha, Oct. 2019 15min
- [Conference talk] German eROSITA Consortium meeting, Ringberg, Oct. 2018, 15min

TUTORIALS:

- X-ray astronomy, Uni-Bonn, 2018
- high-energy astrophysics, NAOC, 2015
- high-energy astrophysics, PKU, 2015

HONORS AND AWARDS:

- 2016, National scholarship of China
- 2016, CAS-DAAD scholarship
- 2014/2015, NAOC-AMD scholarship
- Outstanding member of student union at College and Graduate school
- 2010, NAOC Scholarships
- 2009, National Encouragement scholarship
- 2007-2011, First scholarship three times and Second one for four times in Shandong University at Weihai